

Set 11

Product Category Analysis
Dataset: Product Categories
Category Sales (in \$)
Electronics 50000
Clothing 35000
Appliances 40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages

and title the chart.

2. Build a table to display the sales data for each product category. Label the table elements.

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive

exploration of product category data.

4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

CODE:

```
import matplotlib.pyplot as plt

categories = ['Electronics', 'Clothing', 'Appliances']

sales = [50000, 35000, 40000]
```

1. Funnel Chart

```
plt.figure(figsize=(8, 5))
```

```
max_sales = max(sales)
```

```
for i in range(len(categories)):
```

```
    width = sales[i] / max_sales
```

```
    plt.barh(
```

```
        categories[i],
```

```
        width,
```

```
        height=0.6
```

```
    )
```

```
    plt.text(
```

```
        width / 2,
```

```
        i,
```

```
        "$" + str(sales[i]),
```

```
        ha='center',
```

```
        va='center'
```

```
    )
```

```
plt.xlabel("Relative Sales")
```

```
plt.ylabel("Product Category")
```

```
plt.title("Product Category Sales Funnel")
```

```
plt.gca().invert_yaxis()
```

```
plt.show()
```

```
# 2. Sales Table
```

```
print("\nPRODUCT SALES TABLE")
```

```
print("-" * 40)
```

```
print("Category\t\tSales ($)")
```

```
print("-" * 40)
```

```
for i in range(3):
```

```
    print(categories[i], "\t\t", sales[i])
```

```
print("-" * 40)
```

```
# 3. Pie Chart
```

```
plt.figure(figsize=(7, 7))
```

```
plt.pie(
```

```
    sales,
```

```
    labels=categories,
```

```
    autopct='%1.1f%%',
```

```
    startangle=90
```

```
)
```

```
plt.title("Sales Distribution by Product Category")
```

```
plt.show()
```

OUTPUT:

PRODUCT SALES TABLE

Category	Sales (\$)
Electronics	50000
Clothing	35000
Appliances	40000



